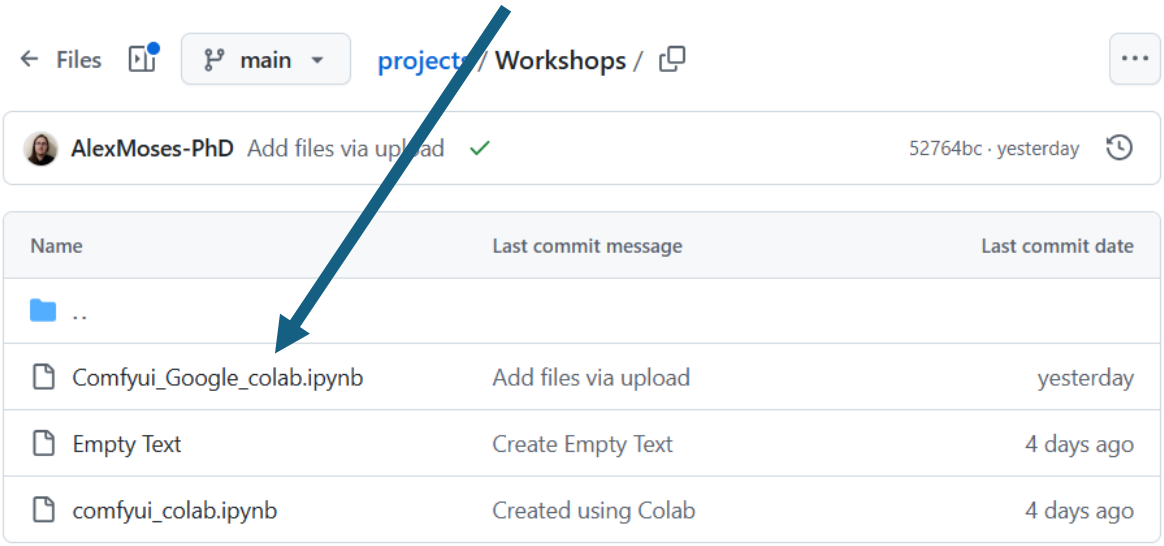
How to setup a basic Art Diffusion Environment

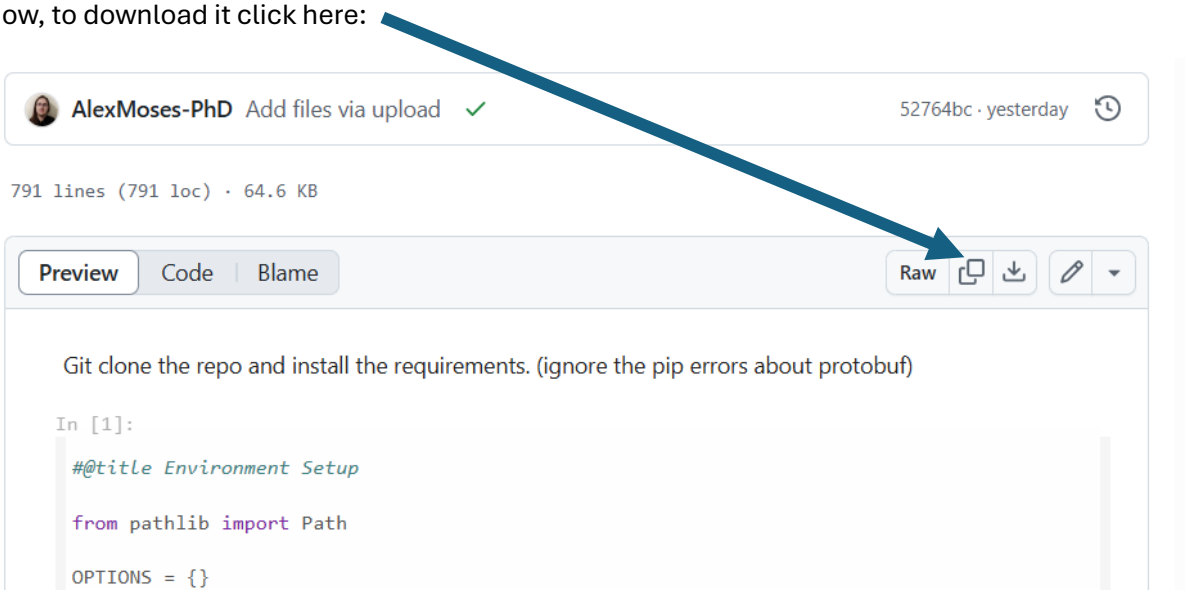
Step 1: Download the notebook

Download the python notebook file from the Warwick AI Github, click the link below: https://github.com/WarwickAI/projects/tree/main/Workshops .

Then click on the file Comfyui_Google_colab.ipynb,



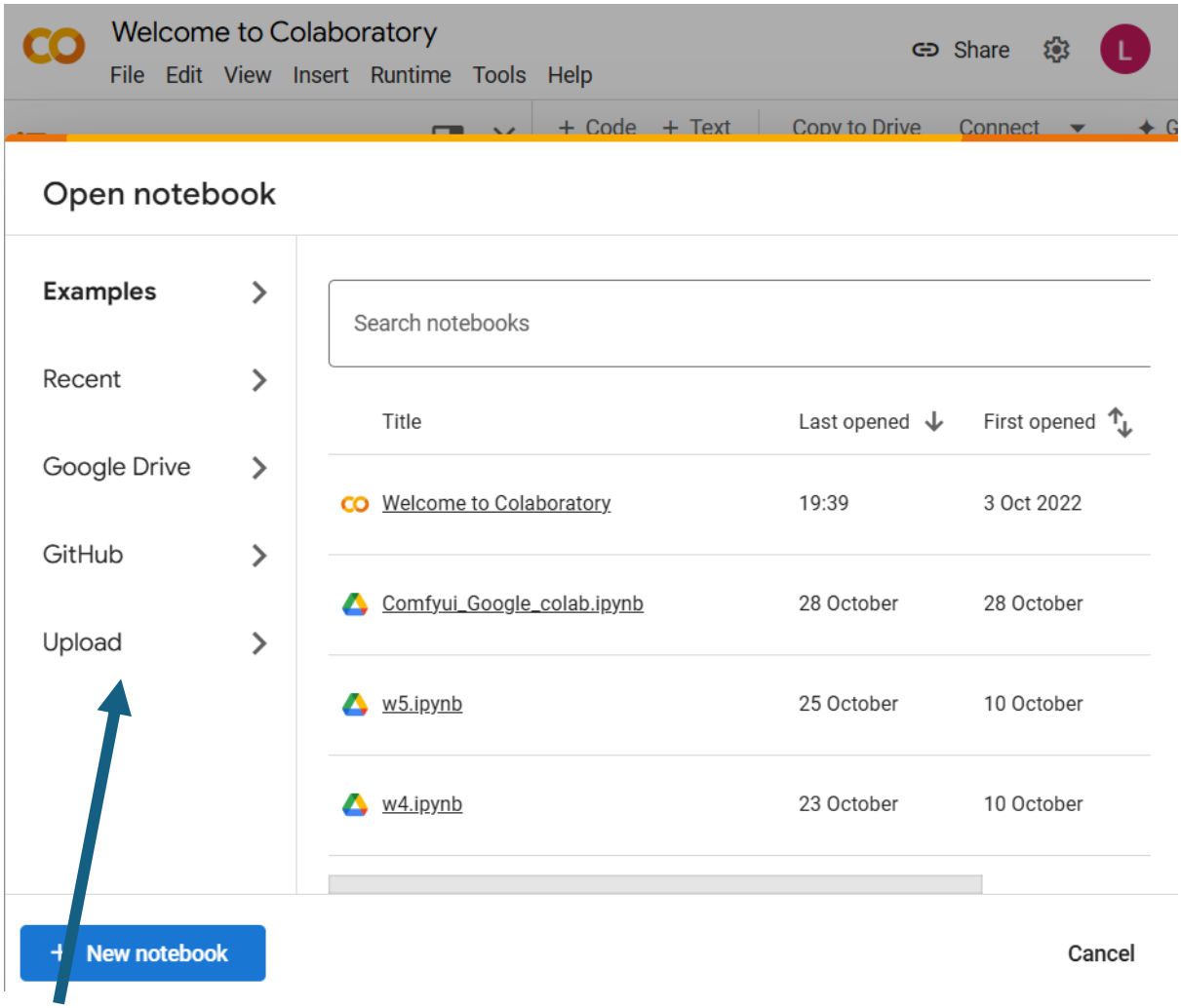
Now, to download it click here:



And save it anywhere / just leave it in your downloads folder.

Step 2: Launch it in Google Colab

Google Colab is a web-based IDE for python which is free to use and provides you with access to some hardware. If you haven't used it before, just type into google, "google colab" and click on the first link. It might ask you to log in – please do, this is just using your google account. Then you should be looking at a page like so:

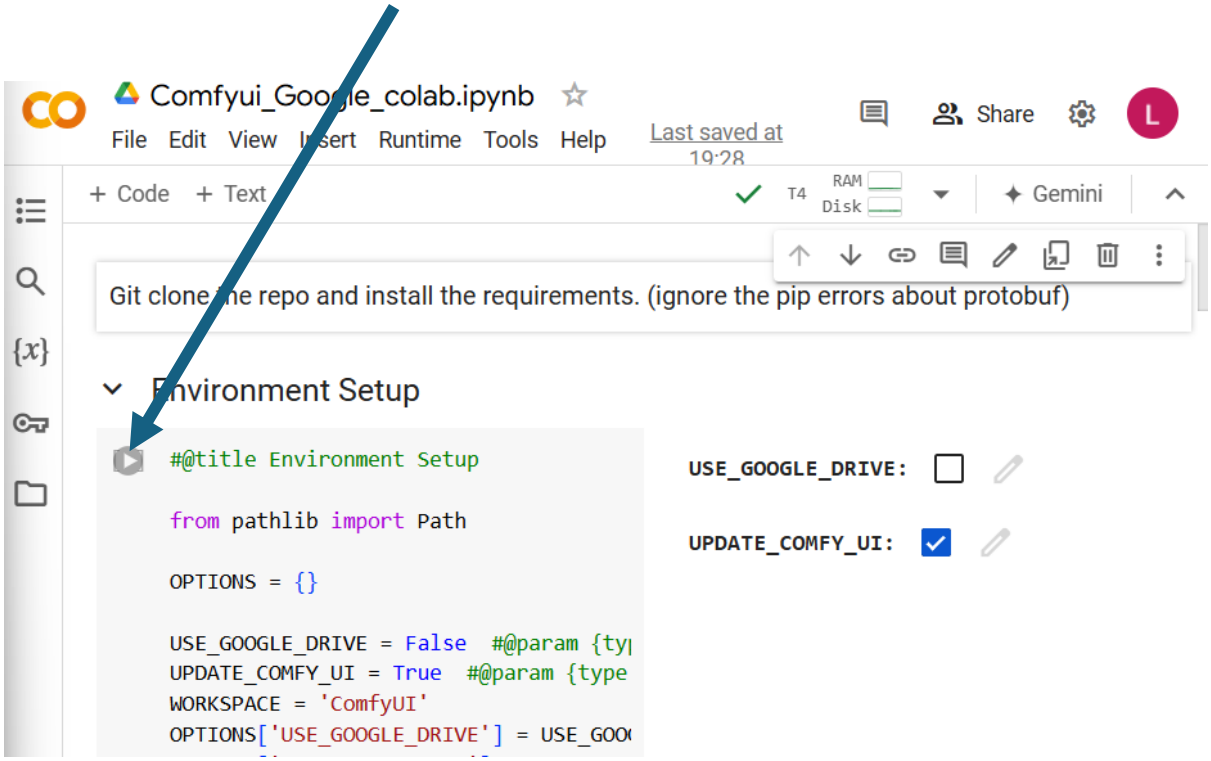


Click upload and choose the notebook file you downloaded in Step 1.

Step 3: Run the notebook

We have written this notebook to setup the diffusion model software and download relevant diffusion models for you. These models will download into the Google Colab environment, so don't worry none of it is actually going onto your computer/laptop.

First, click the first play button under environment setup.



!Important! WAIT for the play button to stop spinning before moving on, it might take up to a minute. This little green tick will appear when it is done.



Now scroll down past a lot of weird text (don't worry about it) until you get to the checkpoints box shown below:

```
# Checkpoints

### SDXL
!wget -c https://civitai.com/api/download/models/782002 -O ./models/checkpoints/ju

# VAE
!wget -c https://huggingface.co/stabilityai/sdxl-vae/blob/main/sdxl_vae.safetensor

# Loras
!wget -c https://civitai.com/api/download/models/135867 -O ./models/loras/add-deta

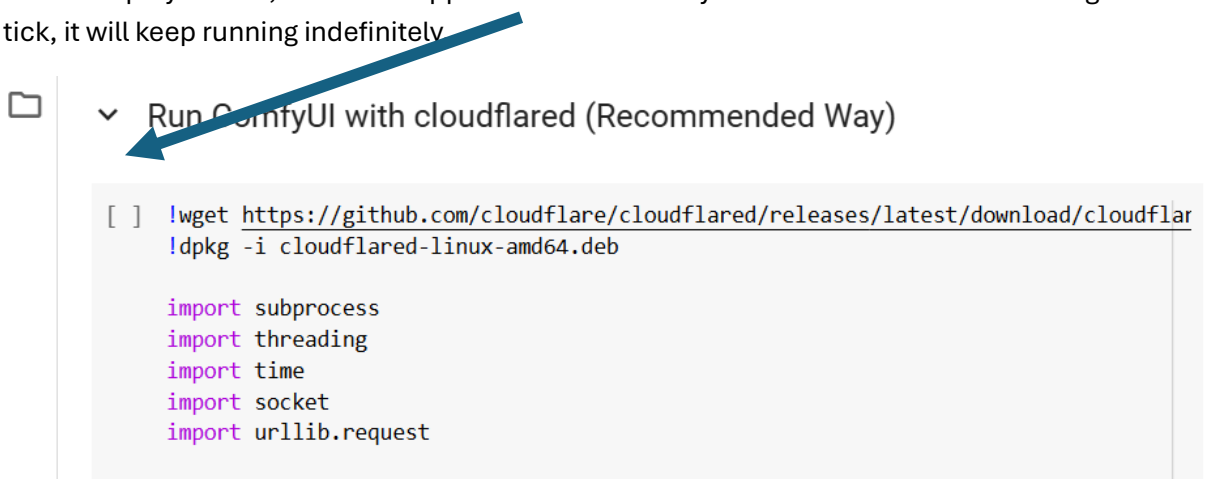
# ControlNet SDXL
!wget -c https://huggingface.co/diffusers/controlnet-depth-sdxl-1.0/resolve/main/c

# Controlnet Preprocessor nodes by Fannovel16
!cd custom_nodes && git clone https://github.com/ltdrdata/ComfyUI-Manager.git

# ESRGAN upscale model
```

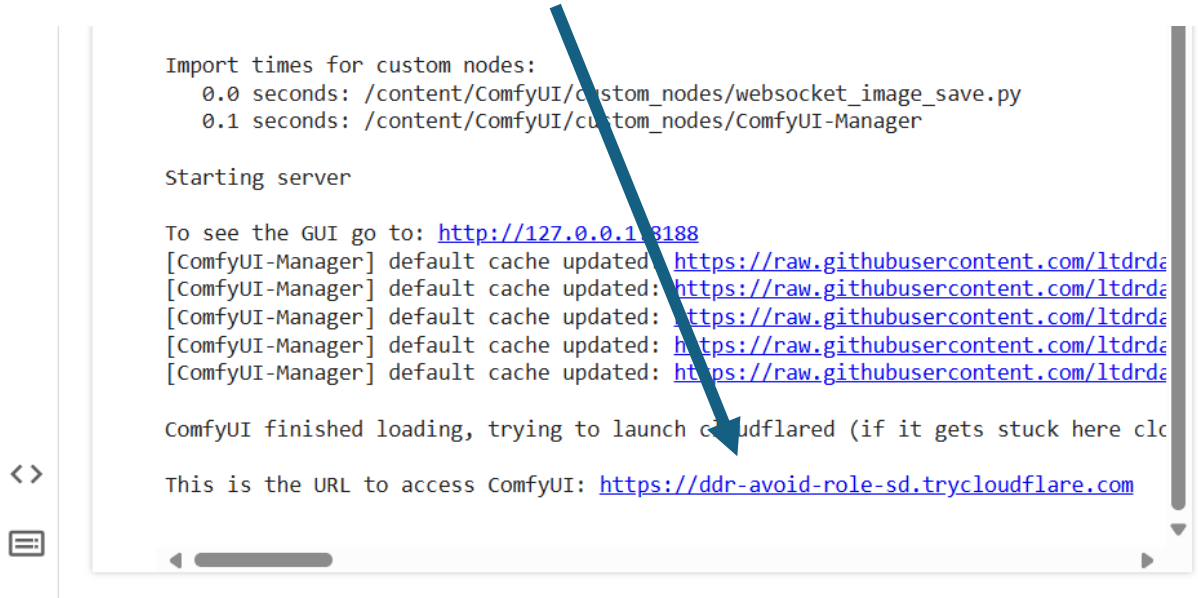
Again, press play, and WAIT for the green tick to appear. This one can take around 5 minutes to complete so maybe go make a cup of tea or something.

Finally, scroll down again past lots of text until you find the Run ComfyUI with cloudflared box. Press the play button, which will appear here. This time you do not need to wait for the green tick, it will keep running indefinitely.



Wait a bit and scroll down past this box, and wait for all the text production to stop until you see the following line:

This is the URL to access ComfyUI: blahblahsomelink



Your link will look different to mine here which is completely normal. Click it and be patient as the UI loads.

See the next page for the UI ->

Step 4: Have Fun!

The software we are using is called ComfyUI (its documentation isn't fantastic but if you want to learn more about it there is a good basic tutorial/guide here: https://stable-diffusion-art.com/comfyui/)

If you want to find out more about diffusion models and what some of the other settings in the UI mean. How does Stable Diffusion work? is an excellent article explaining at a not-to-technical level.

If you're ever struggling we have a workshop in Term 1 and will possible run another in Term 2 going into the more advanced workflows and showing you how to really control the images produced. You can also come to CodeNights or drop a message in #general or #ai-art-workshop channels in the discord.

Don't forget to show us any cool (or cursed) art you make in the #ai-art-workshop channel on discord!

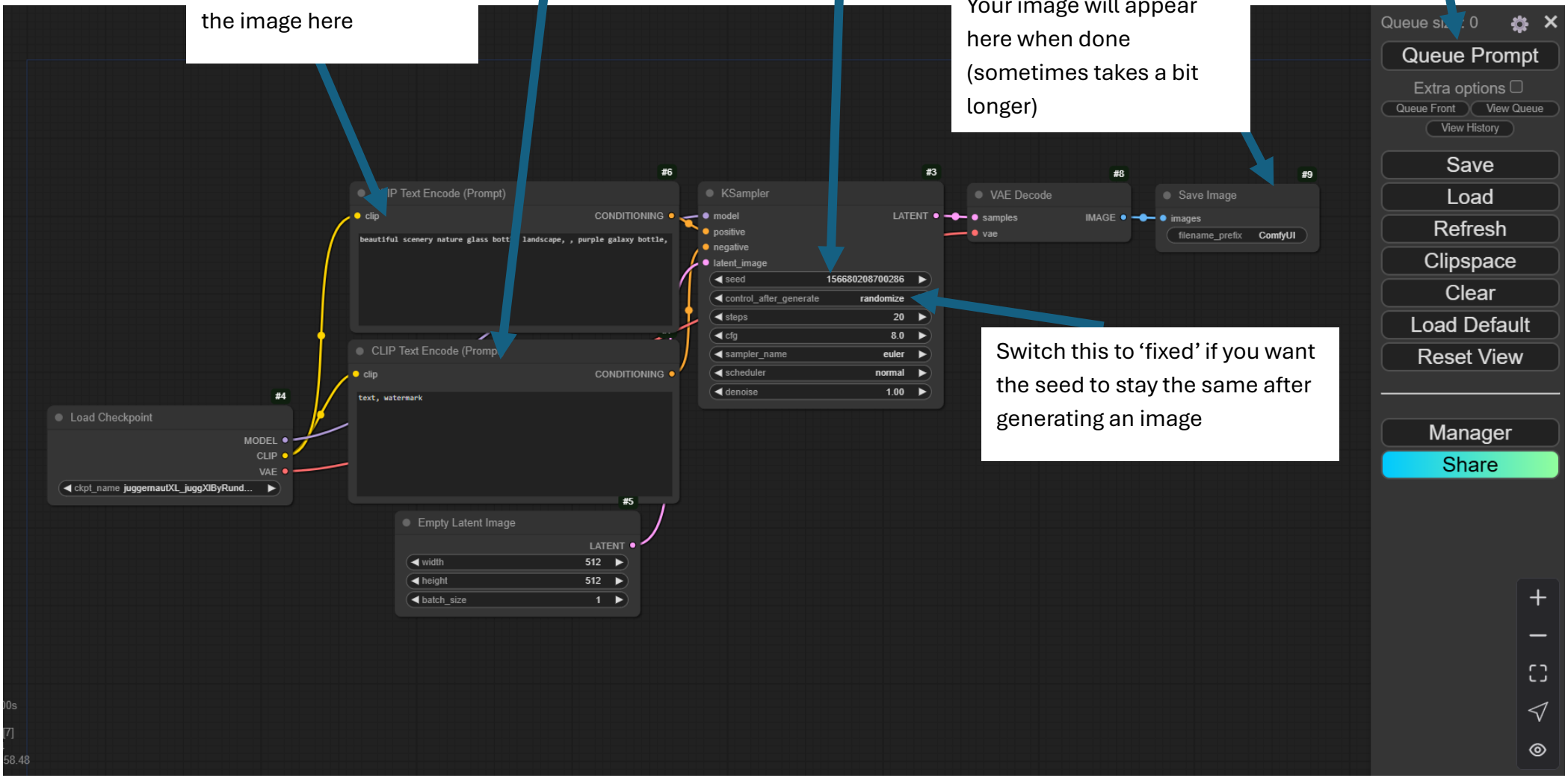
Type things you want in the image here

Type things you <u>don't</u> want in the image here

Changing this number will completely change your output with the same prompt

Your image will appear here when done (sometimes takes a bit longer)

Click here to generate an image!



Switch this to 'fixed' if you want the seed to stay the same after generating an image